

A Splined Outer Mould Line for the SV-1

The same vehicle, re-sized with its nose and boattail revolved from B-splines inside nTop

Prepared with nTop Automate 5.53.2 / 5.54.0 and SUAVE 2.5.2. All geometry is authored programmatically. Every number is read from the run artefacts under runs/SV-1_spline. The tangent-ogive result in examples/SV-1 is the baseline.

Summary

This report describes one change to a coupled sizing loop. The outer mould line of the SV-1 rocket is now a true revolved B-spline. It was a tangent ogive and a conical boattail. Nothing else changed. The requirements, the physics modules, the sizing search and the trade-study axes are the same.

The loop found a design that meets all 10 of 10 constraints. It weighs 551.3 kg. It flies 194.0 km. It arrives at Mach 1.68. The search used 60 evaluations with real nTop geometry in the loop.

Three results are worth more than the mass number. First, the sizer declined the drag optimum: nose_blend converged to 0.70, not 1.0. The shape trade has an interior optimum. Second, the drag saving was not free: peak dynamic pressure rose against a limit that was already close. Third, the coupling decided feasibility. The same design vector with closed-form geometry is lighter, flies further, and violates two constraints. Only the measured geometry finds the feasible answer.

Quantity	Tangent ogive	Spline	Change
Launch mass [kg]	554.34	551.27	-3.07
Range [km]	189.48	194.00	+4.52
Impact Mach [-]	1.659	1.675	+0.017
Peak dynamic pressure [kPa]	195.10	199.11	+4.01
Static margin [cal]	not recorded in the example	1.592	

Table 1. The headline comparison. Both designs meet every constraint. The tangent-ogive figures come from examples/SV-1/01_design/point_ntop.json. The spline figures come from runs/SV-1_spline/converged/point_ntop.json.

1. What changed, and what did not

Only the shape family of the outer mould line changed. The nose is a cubic B-spline with 9 control points. The boattail is a splined contraction. Two new scalars let the sizing search move between the shape the ogive gives and the shape linear theory prefers.

Design variable	Meaning	Bound	Converged
nose_shape	shape family of the forebody	ogive or spline	spline
nose_blend	0 reproduces the ogive, 1 is the drag optimum	0.0 to 1.0	0.70
boattail_shape	shape family of the aft contraction	cone or spline	spline
boattail_blend	0 is the straight cone, 1 is the full curve	0.0 to 1.0	0.35
n_ctrl_oml	control points per spline	fixed	9

Table 2. The design variables the shape change adds. Every other entry in the design vector keeps the meaning it had in the tangent-ogive study.

One thing did NOT change, and it matters. The drag calibration factor stays where it was. It is applied at the loop boundary, in sizing/loop.py, through CalibratedAero. Its value here is 1.171. The aerodynamic model still reports what its physics gives. The loop still owns the correction.

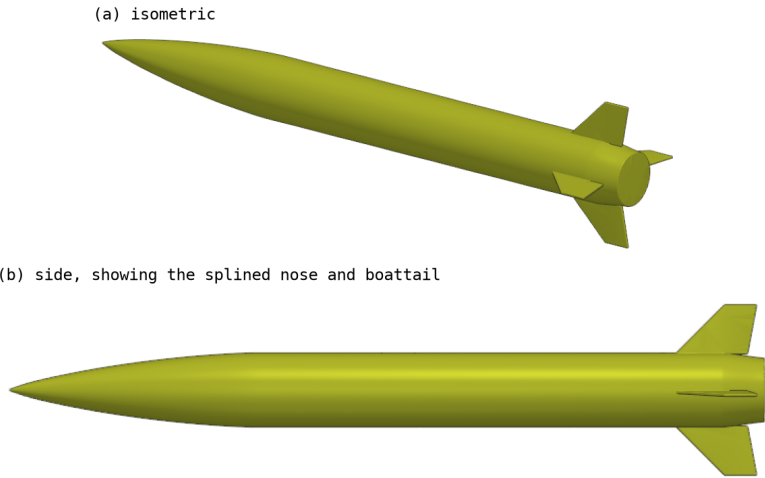


Figure 1. The converged spline vehicle, rendered from the exported STL with a locked camera. The mesh is a picture only. Every number in this report comes from the notebook's own mass properties, which beat the mesh by about sixteen times on the smoke sphere.

2. Why the shape can change anything at all

Before this work no drag model here could tell one nose from another at fixed fineness. The forebody wave-drag term is the Bonney correlation, a function of length over diameter alone. The Sears-Haack cross-check is a function of diameter over length alone. A spline study run against that model would have reported no change, for the wrong reason.

rocketgen/sizing/wavedrag.py supplies the missing sensitivity. It computes the linearised slender-body wave drag of a given area distribution through the Glauert series. It then reports a dimensionless **RATIO** against the tangent ogive of the same fineness. The ratio multiplies the Bonney value. So the calibrated level and the Mach dependence stay with the correlation that was validated against 23 Basic Finner free-flight shots. Only the shape effect comes from linear theory.

At the ogive control values the ratio is exactly 1.0. The test suite asserts that the drag coefficient is then reproduced bit for bit, with an equality check rather than a tolerance. A shape model that changes the answer at the baseline shape would be a defect, not a feature.

Check	Closed form	Model	Relative residual
Sears-Haack body drag	$128 V^2 / (\pi L^4)$	0.968990	4.5e-5
von Karman ogive drag	series against the closed form	0.00267257	1.3e-5
von Karman C_D on base area	$(d / L)^2$	0.02777813	1.3e-5
Optimum shape factor	$4 / \pi$	1.27332596	6.8e-5

Table 3. The wave-drag model against exact closed forms that come from outside this repository. Each test asserts a tolerance of $1.0e-4$. The residuals above are the values the code achieves, measured for this report rather than transcribed from the test file. All four are between $1.3e-5$ and $6.8e-5$.

Stations in the check table	Relative residual
501	$1.7e-2$
1001	$2.6e-3$
2001	$2.0e-4$
4001	$4.5e-5$
8001	$1.7e-5$

Table 4. The Sears-Haack residual belongs to the CHECK TABLE, not to the model. The Sears-Haack profile has an infinite slope at both ends, so the central-difference derivative converges slowly on it. Refining the table drives the residual down by about a factor of three per doubling. A single number could not have said this.

Two further checks support the series itself. The Glauert series constant of π over four was compared against direct double integration of the same integral. The deleted diagonal converges slowly, so what is measured is convergence onto the series: the residual falls from $4.8e-2$ at 250 points to $1.5e-2$ at 1000. Separately, the von Karman ogive was confirmed as the constrained optimum rather than assumed to be. Adding any higher Glauert mode at fixed base area raises the drag in all 4 perturbations tried.

Shape	Glauert shape factor	Over the bound
Tangent ogive	1.49028	1.17038
Optimal 9-point spline	1.30330	1.02354
von Karman ogive, the bound	1.27333	1.00000

Table 5. The gap the spline exists to collect. A 9-point spline recovers 86.2 percent of the distance from the tangent ogive to the theoretical bound. The ogive penalty is nearly independent of fineness: it moves only from 1.1761 to 1.1679 over nose fineness 2.0 to 5.0.

The slender-body wave-drag ratio. It multiplies the calibrated Bonney correlation, so only the SHAPE effect comes from linear theory.
Source: runs/SV-1_spline/figures/evidence.json, section wavedrag.

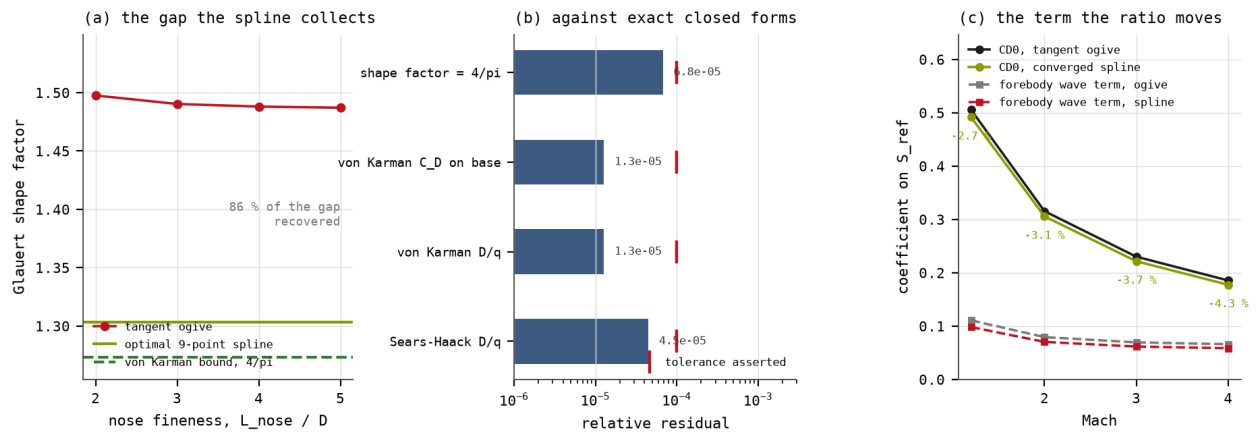


Figure 2. The wave-drag model. Panel (a) is the gap between the tangent ogive and the bound. Panel (b) is every closed-form residual against the tolerance its test asserts. Panel (c) is the drag build-up the ratio moves.

Mach	Wave share of CD0, ogive	Change in wave term	Change in CD0
1.2	21.9 %	-11.37 %	-2.70 %
2.0	25.1 %	-11.37 %	-3.11 %
3.0	30.2 %	-11.37 %	-3.71 %
4.0	35.6 %	-11.37 %	-4.35 %

Table 6. What the shape does on this vehicle, at 12 km and 2 degrees of angle of attack. The forebody wave term is a rising share of zero-lift drag with Mach number. The converged spline cuts that term by a constant 11.37 percent, because the ratio is Mach-independent at this order. The effect on total zero-lift drag therefore grows with Mach number.

3. How the spline is built in nTop

nTop revolves the spline itself. There is no chord polygon and no sampling error. The notebook computes the control points from live inputs, builds a profile from a list of curves, and revolves that profile. The chain is four blocks:

```
spline_by_control_points<list<point>,integer>[5.20.0]
core.list<curve_interface>
profile_from_curves<list<curve_interface>,vector>[5.20.0]
revolve<new_profile,axis,real>[5.20.0]
```

The profile is a curve LIST. It mixes spline segments with two_point_line segments. So the cylinder, the base disc and the return along the axis stay exactly straight, and the corners stay sharp. The axial fractions of the control points are the Greville abscissae. That makes the axial station an exact linear function of the spline parameter, and it keeps the radius a spline in the axial station.

None of those four blocks appears in the vendored block universe, and neither do the types they need. The universe is incomplete, and its silence is not evidence. An earlier attempt sampled the spline into a chord polygon because the universe listed no route from a curve to a revolvable profile. That route exists. All four blocks go through the raw-block escape hatch. docs/NTOP_NOTES.md section 25 records the four encoding traps.

Measurement	nTop	Exact integral of the same spline	Relative error
Outer mould line volume [m^3]	0.287923	0.287941	-0.0064 %
Body wetted area [m^2]	3.50319	3.50267	+0.0148 %

Table 7. The measured solid against the exact integral of the B-spline that built it. Because nTop revolves the spline rather than a polygon, the closed form is an exact description of the solid and not an approximation of it. There is no discretisation error available to hide behind. The volume splits as 0.058686 nose, 0.212627 cylinder and 0.016629 boattail, all in cubic metres.

Notebook	Recipe body nodes	Notebook inputs
SV-1, splined outer mould line	317	37
SV-1, tangent ogive	405	19

Table 8. Notebook size, counted as nodes in the recipe body. The spline notebook is the smaller of the two. A revolved spline replaces the sampled polygon and its supporting arithmetic, and it takes twice as many inputs to carry the control values. Measuring the converged geometry took 133.1 s of wall time in this run.

Quantity	nTop	Closed form	Relative error
volume_total	0.287923	0.287937	-0.005 %
area_wetted_body	3.503191	3.502633	+0.016 %
area_wetted_fins	0.505578	0.453560	+11.469 %
area_base	0.070686	0.070686	+0.000 %

Table 9. Every measurement at the converged design against its closed form. The fin area is the one large disagreement, and it is expected. The closed form counts two sides of each exposed panel. The nTop plate also has a tip face, two edge faces and the cylindrical root patch the boolean leaves

behind.

4. The sizer declined the drag optimum

nose_blend converged to 0.70, not 1.0. boattail_blend converged to 0.35. The drag optimum was inside the bounds and the search moved away from it. The reason is visible in the sweep. Past about seven tenths of the way to the optimum, the forebody volume the shape gives up, and the aft shift in centre of pressure it causes, cost more than the remaining wave drag saves.

nose_blend	Launch mass [kg]	Range [km]	q_max [kPa]	Static margin [cal]	Feasible	Penalty
0.00	552.101	188.85	193.09	0.9981	no	0.55035
0.50	551.691	192.87	197.83	1.0056	yes	0.50154
0.70	551.271	193.78	198.55	1.0183	yes	0.50116
0.85	551.142	194.31	198.75	1.0269	yes	0.50104
1.00	551.232	194.39	198.22	1.0283	yes	0.50112

Table 10. The nTop-COUPLED blend sweep. Every row rebuilt and re-measured the solid. None of the 5 points failed. This sweep is the evidence behind the interior-optimum claim, because the analytic mass model reads a closed form of the mould line and cannot see a re-measured solid. The search minimises the penalty column, not the launch mass.

Read this row carefully. The lowest penalty of the five measured points is at nose_blend 0.85, not at the 0.70 the search stopped on. The optimum is therefore confirmed as interior, because blend 1.0 is worse than blend 0.85 on both penalty and launch mass. But the search did not reach it. The five penalties span only 0.04931, so the objective is nearly flat in this variable and the search had little to follow. The claim this report makes is that the optimum is interior. The claim it does NOT make is that the converged blend is the optimum.

nose_blend	Shape ratio	Nose volume	Nose wetted	CD0 at M 2	x_cp shift
0.00	1.00000	+0.00 %	+0.00 %	+0.00 %	+0.0 mm
0.20	0.95487	-1.30 %	-0.54 %	-1.22 %	+2.9 mm
0.40	0.91991	-2.58 %	-1.09 %	-2.18 %	+5.8 mm
0.60	0.89503	-3.85 %	-1.63 %	-2.87 %	+8.6 mm
0.70	0.88634	-4.48 %	-1.90 %	-3.11 %	+10.0 mm
0.80	0.88015	-5.11 %	-2.17 %	-3.29 %	+11.4 mm
0.90	0.87644	-5.73 %	-2.43 %	-3.41 %	+12.8 mm
1.00	0.87521	-6.34 %	-2.70 %	-3.45 %	+14.2 mm

Table 11. What each blend costs and buys, at fixed everything else. The shape ratio falls quickly at first and then flattens. The volume given up keeps rising at a steady rate. The centre of pressure keeps moving aft, which is what erodes the static margin. That is the whole trade in three columns. Every profile in the sweep stays monotone, and the steepest one reaches a slope of 0.430, so none of them is an unbuildable nose.

The splined outer mould line. Panels (a) and (b) are the geometry nTop revolves; (c) and (d) are what moving it does.
Source: runs/SV-1_spline/figures/evidence.json and the design vector.

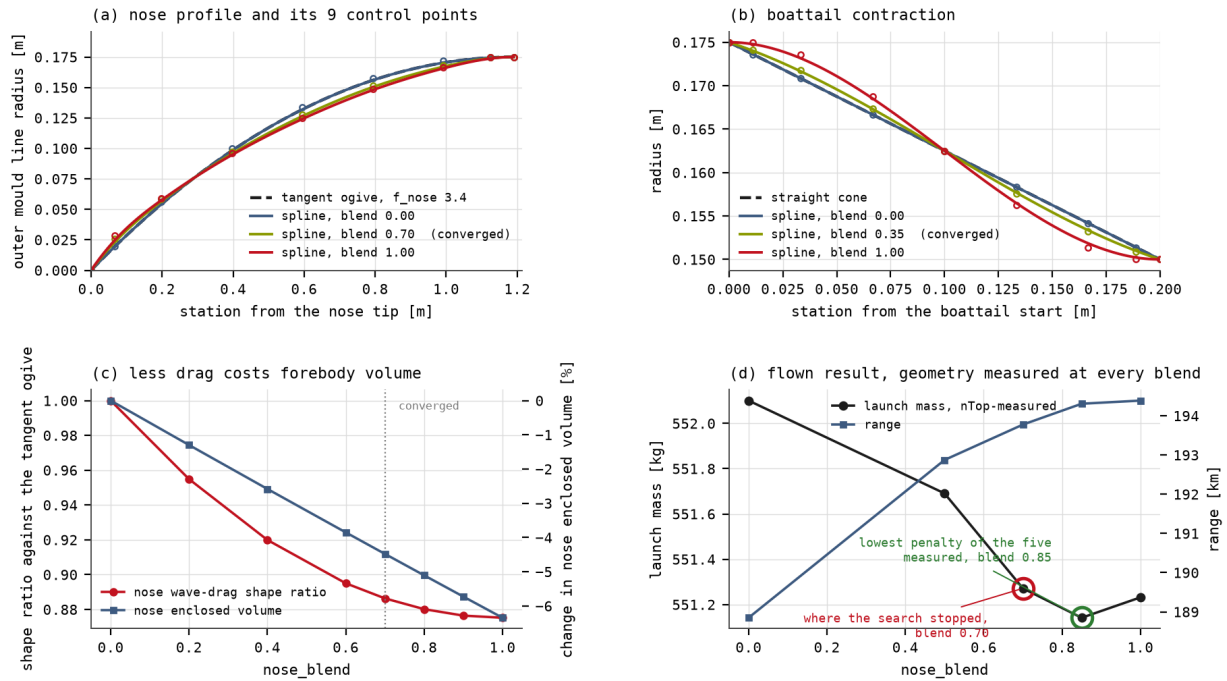


Figure 3. The splined outer mould line. Panels (a) and (b) draw the geometry nTop revolves, with the control points the notebook computes. Panels (c) and (d) are what moving it does. Panel (d) is the coupled sweep, where every point was re-measured.

One honest limitation of the record. The search trace at runs/SV-1_spline/size/search_trace.csv has 60 rows and does not carry nose_blend or boattail_blend. Those variables are searched last, so the trace cannot show the shape search directly. The coupled sweep in Table 10 exists partly to fill that hole.

5. The coupling decides feasibility

The same design vector was evaluated twice. Once with the geometry measured in nTop, and once with the closed-form fallback. The two answers differ by more than the shape change itself.

Quantity	Analytic geometry	nTop measured	Change
Launch mass [kg]	542.57	551.27	+8.71
Range [km]	196.26	194.00	-2.25
Impact Mach [-]	1.683	1.675	-0.008
Peak dynamic pressure [kPa]	201.01	199.11	-1.90
Feasible	NO	yes	

Table 12. The coupling effect at the converged design. The analytic geometry reports a vehicle that is lighter and flies further. It is also infeasible. It violates 2 constraints: R10 static margin, q_max. Only the measured geometry finds the feasible answer. A reader who took the analytic numbers would have shipped a design that does not close.

The mechanism is visible in the mass statement. The measured airframe weighs 38.73 kg. The measured centre of gravity of that structure sits at 2.2533 m from the nose. The closed form does not see the fin roots, the plate edges or the boolean leftovers, so it puts less mass further forward. Less mass raises the peak dynamic pressure. Mass further forward is not enough to hold the static margin.

The centre of gravity of a cruciform body does not measure as exactly on axis. This one is 0.010 mm off in one lateral direction and 0.343 mm off in the other. That is discretisation, not asymmetry. Any test here must use a tolerance and never zero.

The flown mission of the converged design. Solid: geometry measured in nTop. Dashed: the same design vector with the closed-form geometry.
Source: runs/SV-1_spline/converged/point_ntop.json and point_analytic.json.

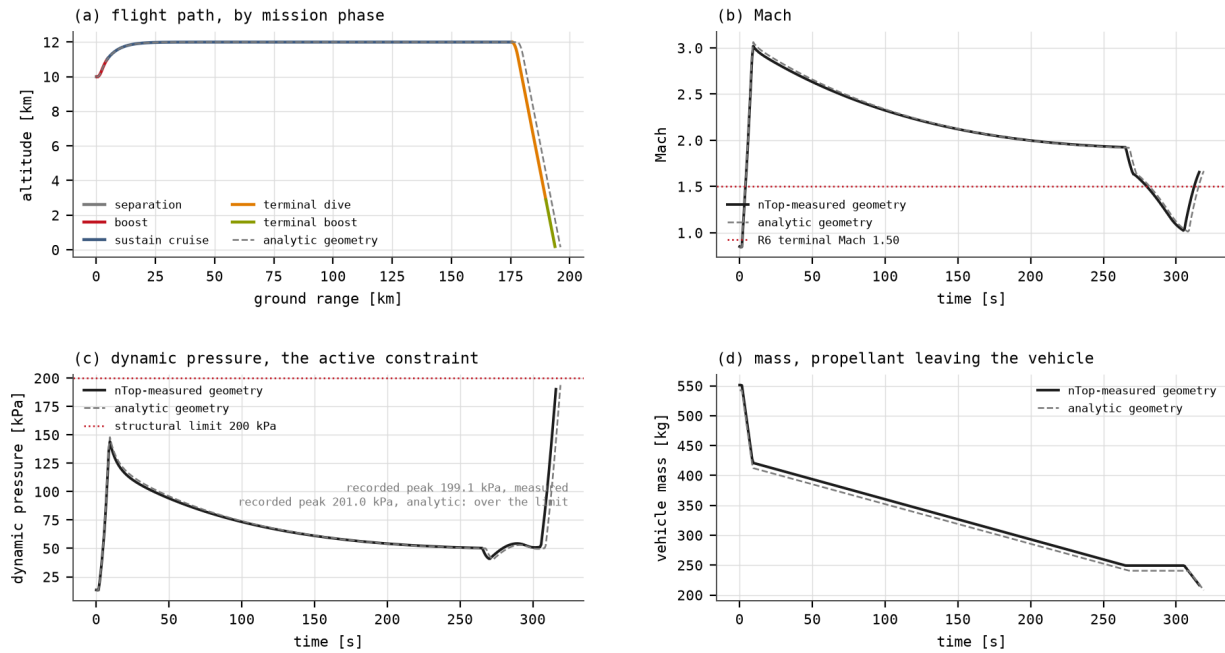


Figure 4. The flown mission. Solid lines are the nTop-measured geometry. Dashed lines are the same design vector with the closed-form geometry. Panel (c) is the one that decides the outcome: the analytic trace crosses the structural limit.

Bars are the nTop-coupled result; the red ticks are the same design vector with analytic geometry. The shaded band marks margins below 5 percent. Source: runs/SV-1_spline/converged/point_ntop.json.

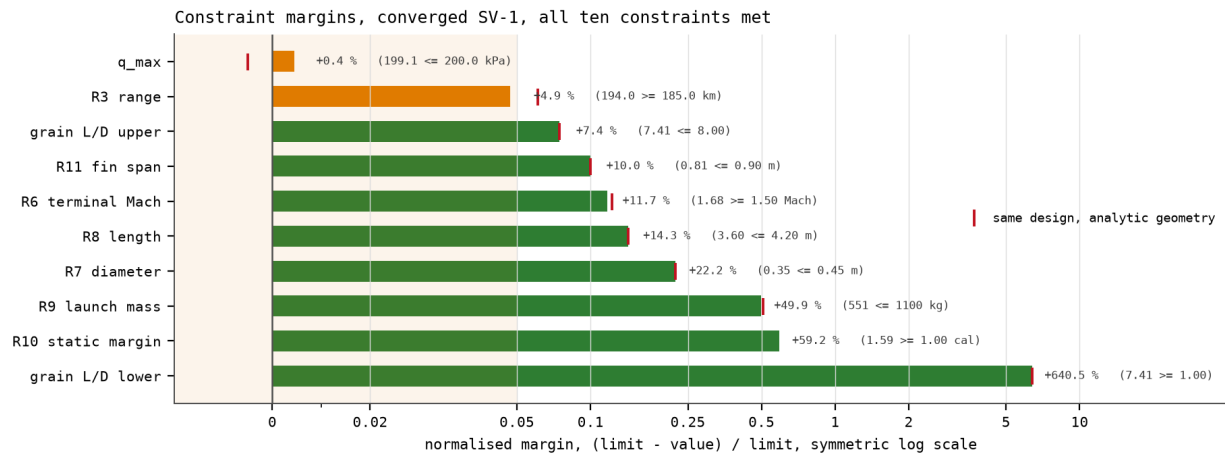


Figure 5. Constraint margins at the converged design. Bars are the coupled result. The red ticks are the same design vector with analytic geometry. Two ticks fall on the wrong side of zero.

6. The converged design

Constraint	Value	Sense	Limit	Met
R3 range	194,003.170	>=	185,000.000	yes
R6 terminal Mach	1.675	>=	1.500	yes
R7 diameter	0.350	<=	0.450	yes
R8 length	3.600	<=	4.200	yes
R9 launch mass	551.271	<=	1,100.000	yes
R10 static margin	1.592	>=	1.000	yes
R11 fin span	0.810	<=	0.900	yes
q_max	199,111.270	<=	200,000.000	yes
grain L/D lower	7.405	>=	1.000	yes
grain L/D upper	7.405	<=	8.000	yes

Table 13. All 10 constraints the loop checks, at the converged design. The dynamic-pressure margin is the smallest, at 0.44 percent. Say how many constraints were checked, not only that they were met: a requirement that is not in this list is not tested.

Mass item	Mass [kg]	Station [m]	Provenance
Warhead	90.00	0.825	requirement
Guidance, seeker, actuation	15.00	0.425	requirement
Radome	13.52	0.535	analytic
Airframe structure and fins	38.73	2.253	ntop_measured
Motor case	5.60	2.250	analytic
Motor insulation	10.51	2.250	analytic
Nozzle assembly	2.85	3.520	correlation
Igniter	1.71	1.150	analytic
Motor hardware not modelled	13.15	2.250	correlation
Propellant, boost	130.00	1.537	analytic
Propellant, sustain	172.00	2.553	analytic
Propellant, terminal	40.00	3.265	analytic
Dry-mass contingency	18.19	1.768	correlation
TOTAL	551.27	1.913	

Table 14. The group-weight statement, with the provenance of every line. Only 7.03 percent of the launch mass is measured in nTop. That is the airframe and the fins, and nothing else. The motor case, the propellant, the warhead and the avionics are charged separately, so the body measured in nTop must be the airframe alone. Double counting here would be silent and large.

Launch mass 551.3 kg, CG 1.913 m from the nose tip, burnout mass 209.3 kg. nTop-measured share 7.0 %.
Source: runs/SV-1_spline/converged/point_ntop.json.

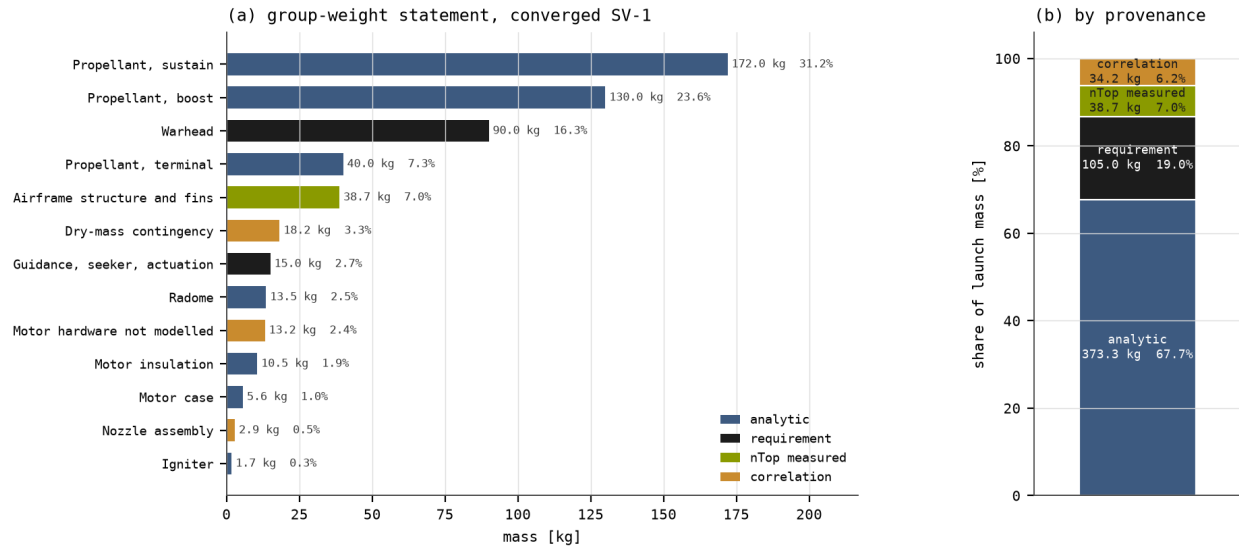


Figure 6. The mass statement by provenance. The measured fraction is small by design. The correlation lines are where the bottom-up model is known to be incomplete.

S(x) from the nTop notebook against the closed-form outer mould line plus plate fins.
Source: runs/SV-1_spline/converged/measurements.json.

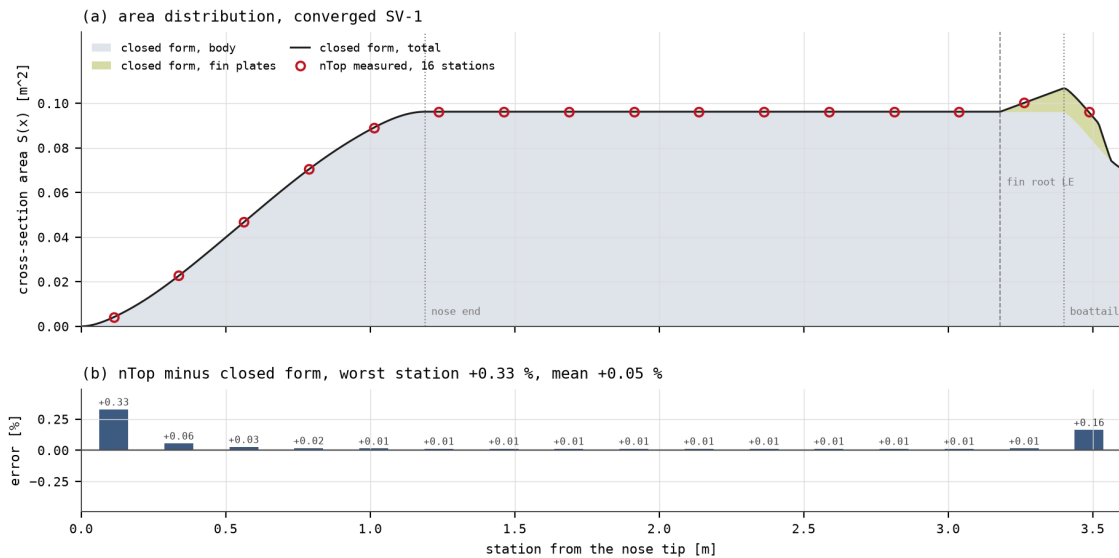


Figure 7. The cross-sectional area distribution measured in nTop, which is the input the slender-body wave-drag model integrates.

7. Trade study

The trade study holds 45 factorial nodes and 40 Latin hypercube samples. That is 85 samples in total. 0 failed to converge. 3 are feasible. Every sample ran with real nTop geometry. A sample that crashed would be recorded as a non-converged row, not dropped: a trade study that drops its failures reports a feasible region that is too large.

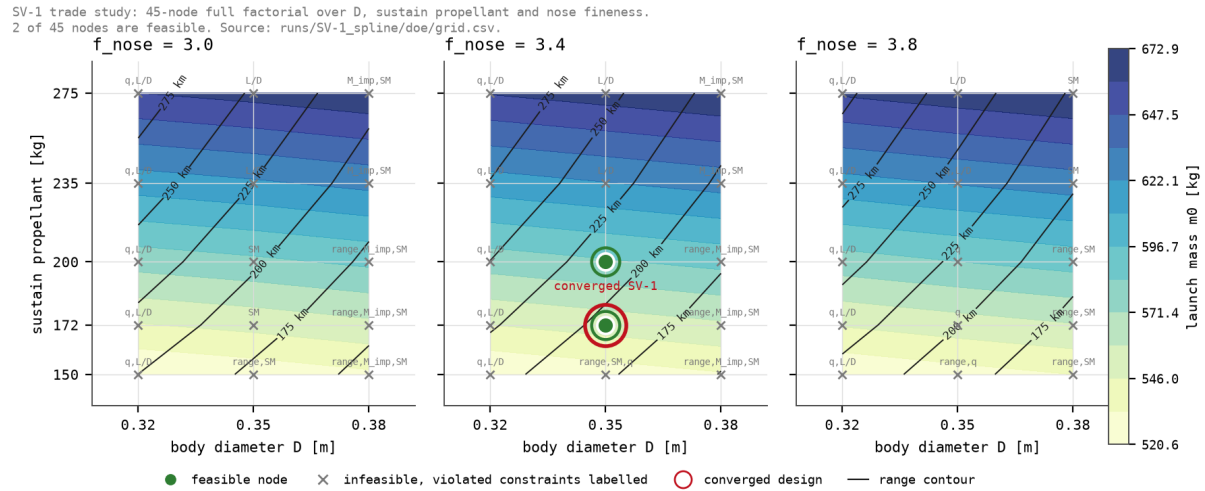


Figure 8. The factorial, drawn as carpet plots. The converged design is a grid node, so the study brackets the answer rather than straddling it. Infeasible nodes carry the names of the constraints they violate.

Variable	m0_kg	range_km	mach_terminal	q_max_kPa
D	+0.13	-0.79	-0.75	-0.66
L_total	+0.14	+0.12	-0.14	-0.11
f_nose	-0.01	-0.00	-0.06	+0.11
m_p_sustain	+0.72	+0.58	-0.03	-0.19
m_p_boost	+0.56	+0.12	-0.28	-0.05
m_p_terminal	+0.01	-0.13	+0.63	+0.40
b_fin	-0.03	-0.18	-0.13	-0.08
c_r_fin	-0.35	-0.20	-0.00	+0.08

Table 15. Spearman rank correlations from the Latin hypercube. Each response has a distinct dominant lever, which is what makes the design tractable. Sustain propellant drives launch mass. Body diameter drives range, impact Mach and peak dynamic pressure, all in the same direction. Terminal propellant is the only strong lever on impact Mach that does not also cost range.

Sensitivity from the 40-sample Latin hypercube over eight variables (0 failed samples). Source: runs/SV-1_spline/dae/sensitivity.json.

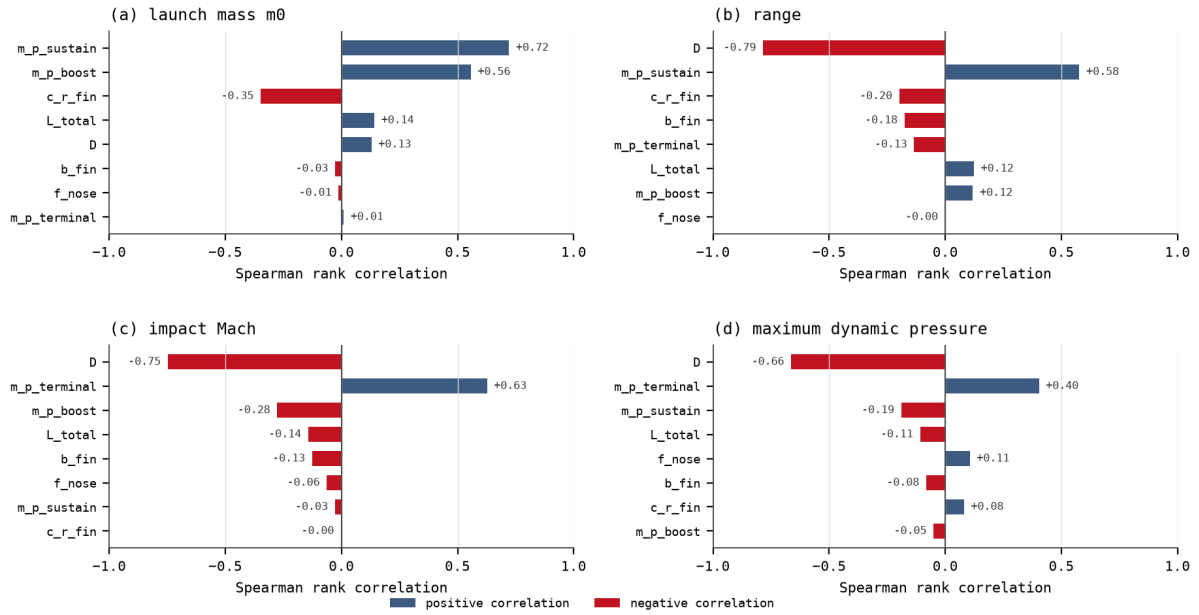


Figure 9. The same correlations, drawn. A short bar means the variable is not a useful lever on that response, whatever it does elsewhere.

8. Verification

The physics under this study is the physics validated for the tangent-ogive SV-1. It is not revalidated here. The residuals below were measured again for this report, so the report quotes what the code achieves rather than the bound its test asserts.

Check	Reference	Relative residual
Vacuum ballistic range	closed-form parabola	2.0e-14
Vacuum apogee	closed form	2.1e-15
Burnout speed	Tsiolkovsky less gravity loss	1.7e-13
Terminal velocity	$\sqrt{2 m g / (\rho S C D)}$	8.7e-10
Specific energy over 100 s	conservation	6.5e-15

Table 16. The trajectory integrator against closed forms. Halving the time step cuts the range error by 15.93, 16.01, 15.76, against the factor of 16 a fourth-order method must give. The order check uses a 45 degree vacuum parabola. A drag-free vertical climb would not work: its acceleration is constant, so the integrator is exact on it and the ratio means nothing.

Quantity	Shots used	Mean bias	Worst shot
Zero-lift drag	17	-14.6 %	22.9 %
Normal-force slope	13	-10.7 %	29.0 %
Centre of pressure	13	+2.0 %	11.4 %

Table 17. The aerodynamic build-up against 23 Basic Finner free-flight shots from DREV-TM-9703, Table VII. The drag bias is where the calibration factor of 1.171 comes from. The calibration is applied at the loop boundary and never inside the aerodynamic model.

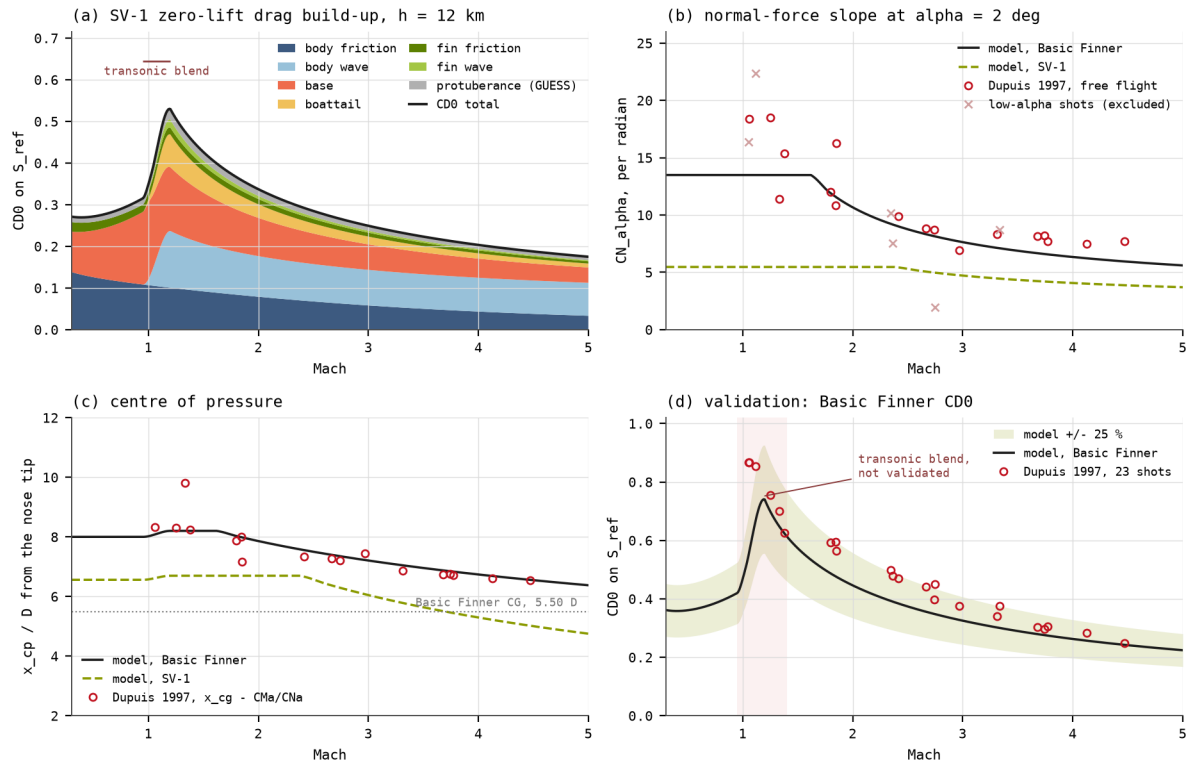


Figure 10. The aerodynamic build-up and its validation. Panels (a) and (b) use the default SV-1 design vector, not the converged spline one, because the panel exists to show the build-up and the Basic Finner comparison. The converged spline drag is in Figure 2 panel (c).

9. Limitations

The source registry holds 96 entries at the point this study wrote it. 14 of them are flagged as guesses. Every empirical constant in this repository carries a source string. If the value is a guess, the word GUESS must appear in that string, and a test asserts it. The table below is the complete flagged list, so a guess that hides becomes a guess that ships.

The registry is filled at import time, so it is only complete once every owning module has been imported. The writer of this file imports the aerodynamic, atmospheric, mass, propulsion and trajectory modules explicitly for that reason. A provenance file written before a trajectory was flown once listed 37 sources instead of 70, and under-reported its own limitations by a factor of four.

Flagged source	What it says
aero_nozzle_exit_fraction_GUESS	GUESS. Used only when `area_nozzle_exit` is None and `power_on=True`: the nozzle exit is assumed to occupy $A_e/A_{base} = 0.50$. This is a placeholder for the real value, which WP3's motor model must supply. It changes only the powered-base drag relief.
aero_protuberance_GUESS	GUESS. A single lumped allowance for antennas, umbilicals, fin-root fairings, lugs, joints and surface imperfections, taken as 5 percent of the sum of the other zero-lift components. There is no correlation behind this number. It appears in the breakdown as 'CD_protuberance_GUESS' so it can never be mistaken for a computed quantity.
bulkhead_allowance	GUESS: four ring bulkheads, each a full-diameter disc of 40 percent of wall thickness. Not from a source; a placeholder for joint, rail and lug hardware
case_minimum_gauge	Manufacturing minimum gauge, not a strength requirement. Hoop stress alone gives a sub-millimetre wall at this chamber pressure, which cannot be filament wound or machined. Floors of 1.8 mm (composite, roughly 3 to 4 wound plies) and 1.2 mm (metallic) are taken as engineering minimums. GUESS: no public minimum-gauge standard was found, but without the floor the model returns a motor mass fraction of 0.95, well above the 0.80 to 0.92 band for real tactical motors
ogive_polygon_sampling	Measured, not guessed. The revolved OML is a chord polygon, so its volume is the frustum sum that `rocketgen.sizing.masses._tangent_ogive_volume` computes with the same n. Sweeping n against a 20000-segment reference on the default SV-1 gives a TOTAL OML volume error of -0.106 percent at n=8, -0.026 percent at n=16, -0.0117 percent at n=24, -0.0066 percent at n=32 and -0.0004 percent at n=128. n=24 is used for the outer profile: 85x inside the 1 percent gate in PLAN.md, and small enough to keep the recipe near 300 blocks.
prop.gamma	ASSUMPTION (not tabulated, treat as a guess inside a physically bounded band): ratio of specific heats of the combustion products taken as 1.20, the value normally used for aluminized AP/HTPB exhaust. Cross-checked: with $\gamma = 1.20$ the cited Isp (262.5 s) and flame temperature (3440 K) imply a mean exhaust molar mass of about 26 kg/kmol, inside the 25 to 30 kg/kmol band of Sutton Figure 12-3 for HTPB-based composite propellant at 68 atm. A +/- 0.02 change in gamma moves the derived c^* by about 1.5 %.
prop.igniter_mass	GUESS: igniter and safe-arm mass taken as 0.5 % of total propellant mass.
prop.insulation_thickness	GUESS: 3.0 mm of EPDM internal insulation on the case wall and domes. Real insulation is sized from local gas-side heat flux and exposure time, which is out of scope (SPEC.md section 8). Density from MATERIALS['insulation_epdm'].
prop.motor_mass_fraction	CROSS-CHECK CORRELATION, quoted band only, treat the endpoints as a guess: tactical solid rocket motors have propellant mass fractions $m_p / (m_p + m_{inert})$ of roughly 0.80 to 0.91. This band was not verified against a primary source in this session; it is used only to bound the physics-route inert mass, never to replace it silently.

Flagged source	What it says
prop.nozzle_mass_model	GUESS-LEVEL geometric estimate: conical nozzle, 15 deg half angle, 8.0 mm wall, bulk graphite / carbon-carbon density 1800 kg/m ³ (handbook range 1.7 to 1.9 g/cm ³ , not verified against a datasheet in this session), multiplied by 1.5 to allow for the convergent section, throat insert and attach ring. This estimate is known to be optimistic; compare against SOURCES['prop.motor_mass_fraction'].
prop.terminal_pulse	MODELLING CHOICE with one guess: the terminal pulse is a third grain segment in the same case, ignited on command rather than on a timer, which is how real dual-pulse tactical motors work (an inter-pulse bulkhead separates the charges and a second igniter fires the aft one). Its default chamber pressure is 1.0 x DesignVector.p_c, that is the pressure the case is already sized for, so no new number is introduced. GUESS: the inter-pulse bulkhead, its insulation and the second igniter are NOT costed as separate inert mass; the terminal grain is charged only the same 0.5 % igniter fraction as the rest of the motor, so the reported inert mass is optimistic for a pulsed motor. Physical motivation: an unpowered terminal dive is terminal-velocity limited and cannot reach Mach 1.5 at sea level for any dive angle, so thrust in the endgame is the only way to meet SPEC R6.
radome_thickness	GUESS: 6 mm Pyroceram radome wall. No public thickness standard was found for a 0.35 m class radome; scaled from the requirement that wall thickness be an integer number of half-wavelengths at Ku band, which is not modelled here
traj.alpha_limit	MODELLING CHOICE: trim angle of attack is limited to 15 deg. Trim demands beyond that are recorded as an authority shortfall rather than flown, so the sizing loop can see them. 15 deg is a conventional tactical-rocket limit; it was not taken from a specific source in this session, so treat it as a guess.
wave_drag_mach_independence	Leading-order slender-body wave drag carries no Mach number. That is a property of the theory, not an omission. Because it is applied here as a ratio against an ogive evaluated at the same Mach, the Mach dependence of the delivered CD_wave_body is entirely Bonney's. GUESS-FREE but APPROXIMATE: the true shape sensitivity does drift with Mach number, and that drift is not modelled and has not been quantified here.

Table 18. Every flagged source in the registry this study wrote. Two of them belong to the work this report describes: the polygon-sampling entry, and the entry that records the wave-drag ratio as Mach-independent.

What this study does not claim

- The slender-body wave-drag ratio is Mach-independent at this order. The true shape sensitivity drifts with Mach number. That drift is not modelled and it has not been quantified. The registry carries this as a flagged entry.
- The nozzle model is ideal. There is no two-phase loss, no divergence loss, no combustion efficiency and no throat erosion. Real delivered specific impulse for this class runs 3 to 7 percent lower. That penalty is NOT applied, because its magnitude could not be sourced. It is the largest known unquantified optimism in the result.
- The optimum is confirmed interior. The converged blend is not confirmed optimal. The coupled sweep found a lower penalty at a blend the search did not reach.
- The linear theory behind the shape ratio is a small-disturbance theory. It is used here only as a ratio between two shapes of the same fineness, which is the use it is best suited to, but it remains linear theory applied to a body at Mach 4.
- The search trace does not record the two shape variables. The coupled sweep partly fills that hole. It does not replace a traced search.

- The requirements are invented for the demonstration. They do not correspond to any real programme.

Warnings the run raised, and did not swallow

Warning
motor: sustain nozzle: exit pressure 29.5 kPa; the Summerfield criterion predicts flow separation below 2595 m altitude. Reported only, the thrust model does not change.
motor: throat area decreases from boost (4024 mm ²) to sustain (860 mm ²): no insert mechanism shrinks a throat; needs a separate smaller sustainer nozzle or a jettisoned tandem booster
grain: grain length 2.503 m exceeds the available motor bay 2.300 m by 0.203 m
motor inert mass raised from a bottom-up 20.7 kg to the correlation floor 33.8 kg; the bottom-up model is incomplete by 13.2 kg
trajectory: ground impact; sustain phase could not hold constant Mach at constant altitude: the motor was short by up to 5679 N. The trajectory shows what the motor actually delivered.; terminal dive entered at t = 265.4 s; terminal boost lit at t = 305.6 s for 12.7 s; terminal boost was still burning 1.8 s of propellant worth at impact; that propellant was wasted

Table 19. The 5 distinct warnings the converged run raised. They are carried upward on purpose. The motor throat transition needs hardware that does not exist, and the grain is longer than the bay that holds it. Neither is hidden by the constraint list passing.

10. Reproducing this result

```
.venv/Scripts/python.exe -m pytest tests -q
.venv/Scripts/python.exe run_sv1.py --stage smoke --oml spline
.venv/Scripts/python.exe run_sv1.py --stage size --oml spline
.venv/Scripts/python.exe run_sv1.py --stage doe --oml spline
.venv/Scripts/python.exe run_sv1.py --stage converged --oml spline
.venv/Scripts/python.exe -m rocketgen.report.evidence --oml spline
.venv/Scripts/python.exe -m rocketgen.report.fig_oml --oml spline
.venv/Scripts/python.exe -m rocketgen.report.build_report_spline
```

Component	Version or value
Python	3.11.15
numpy	1.26.4
scipy	1.13.1
SUAVE	2.5.2, vendored from github.com/suavecode/SUAVE
nTop Automate	5.53.2 installed, 5.54.0 development build
Drag calibration factor	1.171
Outer mould line family	spline
Sizing evaluations allowed	60
Converged run wall time [s]	158.3
Total study wall time [s]	184.3

Table 20. Environment. numpy and scipy are pinned below their current major versions because SUAVE 2.5.2 does not run on either.

Two toolchain warnings are worth repeating. The nTop command-line tool returns exit code 72 when a block fails, not on success. Widely repeated guidance says the opposite and is wrong. Judge success on the expected files existing and being non-empty. Second, conversion evaluates the notebook, exports included, so a fine mesh tolerance makes conversion itself cost minutes. The pattern is to convert once and run many times, which works because every design variable is a real notebook input.